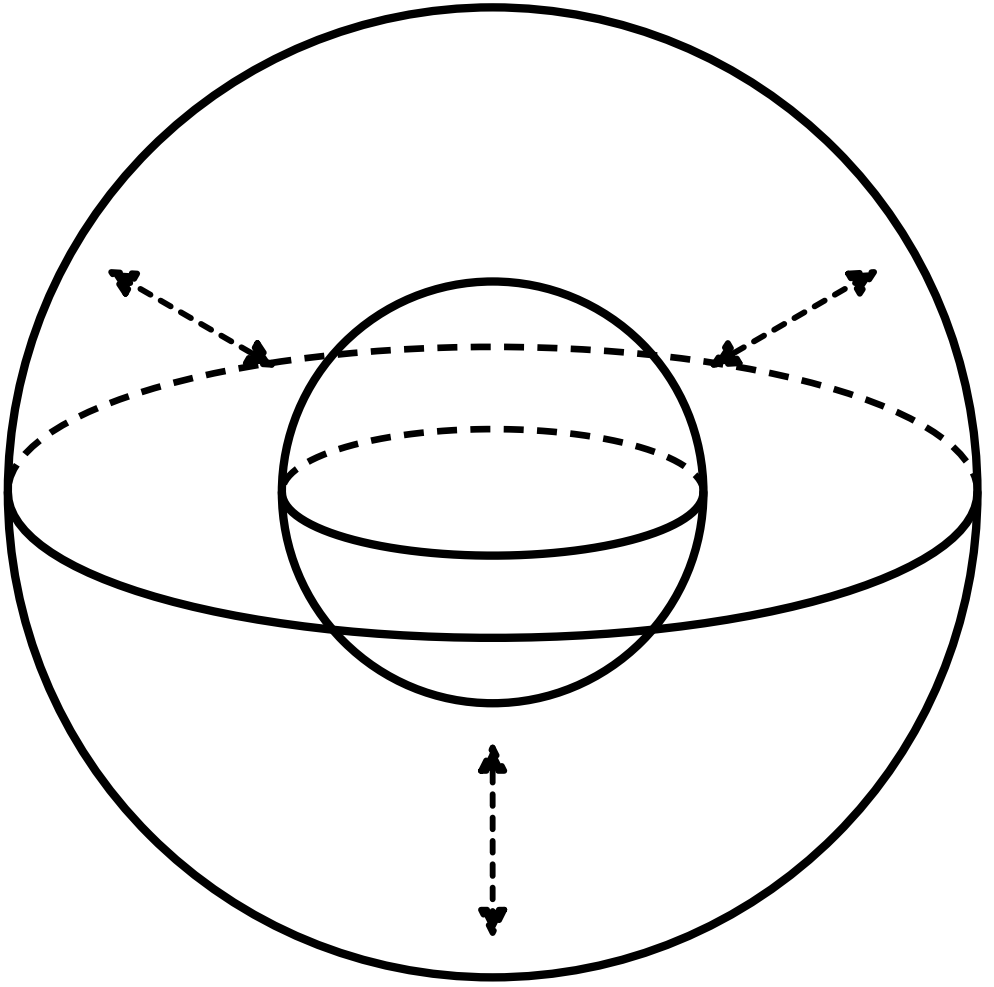

$S^2 \times [0,1]$, inner $\sim$ outer